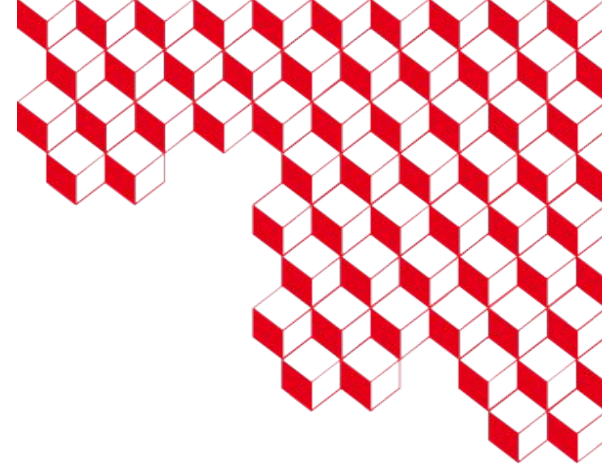




Upgrading a framework doing hallucination correction with a VLM and external tools

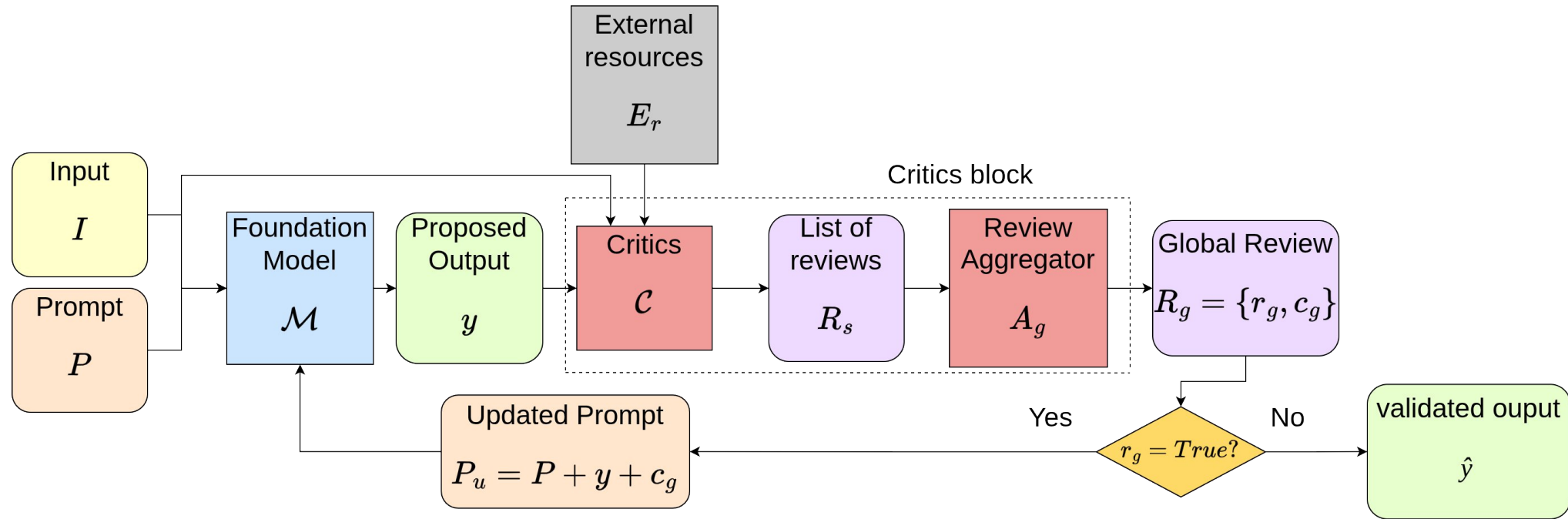
Damien Alouges

Supervisors : Raphaël Lallement, Matteo Morelli
Director : Sébastien Gérard

03/2025-03/2028

09/2026

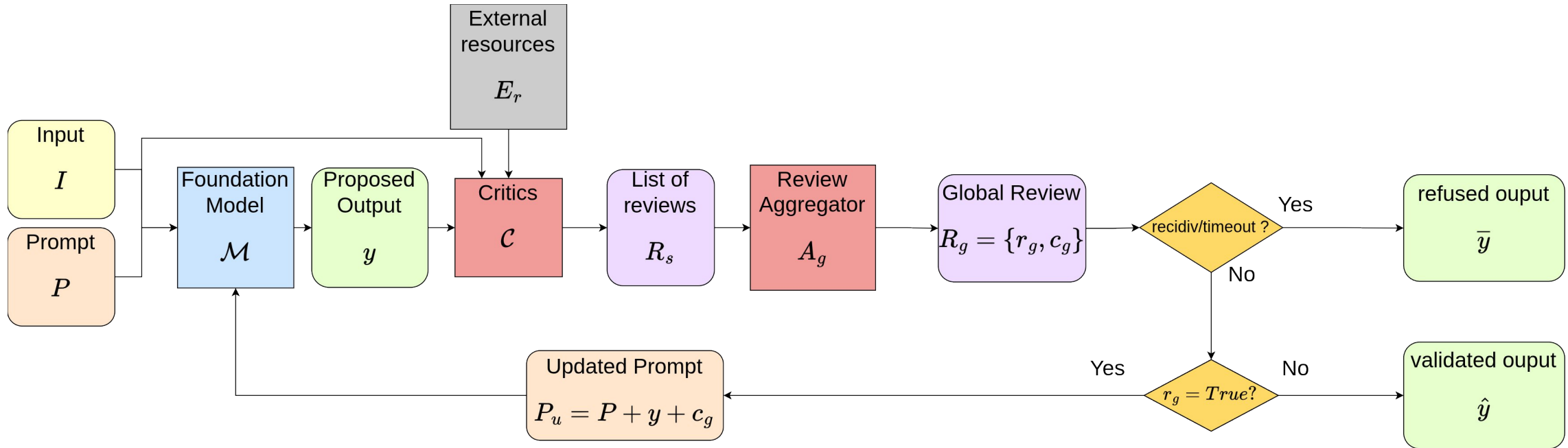
Context : Critics Framework



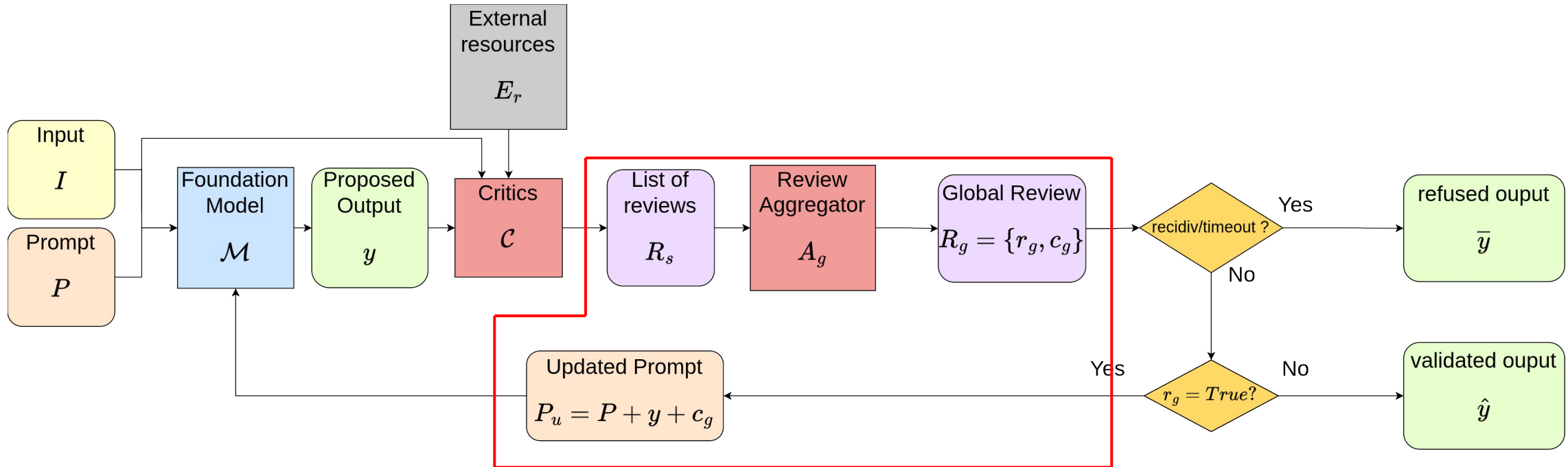
First results :

- reduction of 20% of hallucinations
- observation of recidiv behavior

Critics Framework updated



Experiment goal

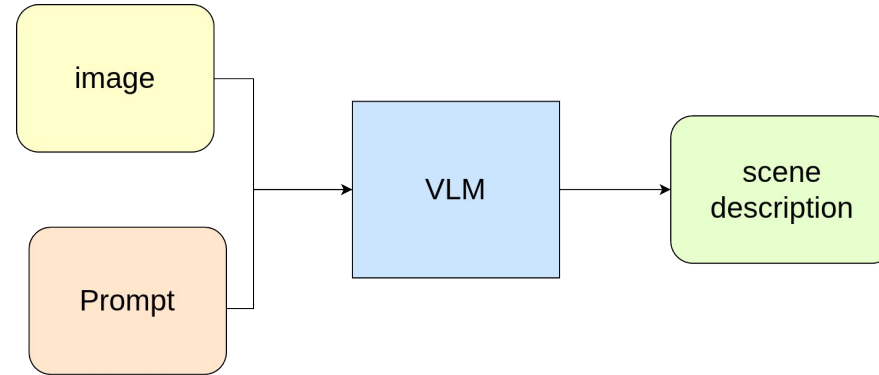


focus on the correction part of the framework -> critics are oracles

Experiment



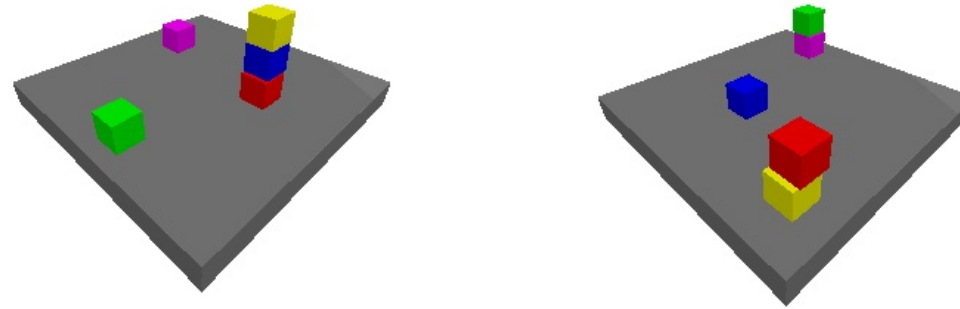
scene description use case :



database

1000 simulated scenes of cubes placed on a surface, for each scene we got :

- 1 images (260x260)
- a ground truth



scene image examples

Configuration tested

- no framework : raw VLM
- framework with oracle critics

review :

- one by one
- stacked : stacked_rev
- all at once : all_rev

updated prompt :

- original sys prompt
- corrector sys prompt : syspu
- no image : no_im

vanilla configuration = oracle critics + one by one + original sys prompt

Metrics



- **generation partial correction rate** (gpcr) : rate of hallucination corrected
$$\text{gpcr} = 1 - \text{hallucinations in tested refused output} / \text{hallucinations in raw output}$$
- **generation full correction rate** (gfcrr) : rate of validated output fully corrected
$$\text{gfcrr} = 1 - \text{fully corrected output} / \text{raw hallucinated output}$$
- **hallucination correction performance rate** (hcpr) : rate of hallucination corrected per loop
$$\text{hcpr} = \text{number of corrected hallucination} / \text{number of loop}$$
- **hallucination creation rate** (hcr) : rate of hallucination created in the refused output
$$\text{hcr} = \text{number of hallucination created} / \text{number of hallucinations in refused output}$$

Results

raw VLM results : 147 hallucinated responses, containing 556 hallucinations

configuration	gpcr ↑	gfcr ↑	hcpr ↑	hcr ↓
vanilla	0.73	0.75	0.17	0.30
stacked	0.72	0.75	0.21	0.14
all_rev	0.92	0.91	0.36	0.07
syspu	0.79	0.79	0.26	0.06
syspu+stacked	0.78	0.79	0.29	0.05
syspu+no_im	0.98	0.99	0.87	0.37
syspu+no_im+all_rev	1.00	1.00	0.96	0.00

limits :

- use case really simple
- critics are oracles
- results on smaller model way worst

Conclusion & futur work



- We observed some configurations more efficient than others
- futur work will focus on the critics
- tests with small models
- tests with harder use cases



Thank you

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